RADAR-SWARM Research-Oriented Simulator for Distributed Radar Networks and Cooperative Drone Swarms

Raj Kokatanur

May 12, 2026

Contents

1 Abstract

RADAR-SWARM is a research-oriented simulation platform for autonomous airspace monitoring, tracking, and interception using a distributed radar network and a cooperative drone swarm. The project combines 3D target dynamics, radar signal processing, multi-sensor fusion, target tracking, distributed swarm intelligence, adaptive sensing geometry, and several quantum-inspired as well as bio-inspired algorithmic concepts. The goal is not only to simulate conventional radar surveillance, but also to study how mobile radar drones can act as both sensing nodes and interceptors within an adaptive, information-driven system.

This report explains every major topic discussed during the project design process, including classical radar architecture, distributed sensing, communication design, information-theoretic planning, synthetic aperture radar, interferometric localization, and quantum-inspired coordination ideas such as time-crystal-inspired synchronization, superposition-inspired multi-hypothesis tracking, and Zeno-inspired track stabilization.

2 Project Vision and Motivation

2.1 Core Vision

The central idea behind RADAR-SWARM is to simulate a system in which multiple drones equipped with radar sensors cooperate with ground radar infrastructure to detect, track, classify, and potentially intercept aerial targets in a three-dimensional environment.

The project moves beyond a traditional single-radar setup and explores a future-oriented concept of **cooperative autonomous sensing**. In this architecture, drones are not only interceptors but also mobile sensing platforms. This means the swarm itself becomes an adaptive radar network whose geometry can change over time.

2.2 Why This Project Matters

The project is valuable because it connects several modern research areas:

- radar signal processing,
- sensor fusion,
- multi-target tracking,
- swarm robotics,
- information-driven decision making,
- quantum-inspired algorithm design,
- bio-inspired distributed coordination.

This makes RADAR-SWARM more than a simple simulation. It becomes a platform for studying how future sensing systems may operate under realistic communication, computation, and uncertainty constraints.

3 System Overview

The final conceptual architecture of RADAR-SWARM can be summarized as follows:

1. A **3D airspace simulator** generates moving targets and drone states.

2. A **distributed radar network** composed of ground radar and drone-mounted radar nodes senses the airspace.

3. A **signal processing layer** performs range-Doppler processing, detection, and possibly waveform correlation.

4. A **fusion and tracking layer** combines multi-sensor measurements and maintains target tracks over time.

5. A **swarm intelligence layer** allocates tasks, maintains formations, and decides on interception or sensing roles.

6. A **communication layer** regulates information flow and bandwidth usage.

7. A **quantum-inspired and bio-inspired coordination layer** provides higher-level strategies for synchronization, search, stabilization, and optimization.

8. An **imaging and advanced sensing layer** may include interferometry, synthetic aperture radar, and classification.

4 3D Environment Model

4.1 Why 3D Was Chosen

A 3D environment was selected over a 2D one because drones and aerial targets operate naturally in three-dimensional space. A full 3D model allows the simulator to capture realistic elevation changes, altitude-based target maneuvers, line-of-sight constraints, and more physically meaningful radar geometry.

4.2 Target State Representation

A standard target state vector in the simulator is:

$$\mathbf{x}_t = [x, y, z, v_x, v_y, v_z]^T \tag{1}$$

where x, y, z denote position and v_x, v_y, v_z denote velocity components.

4.3 Drone State Representation

Each drone may also use a state vector of the same general form:

$$\mathbf{x}_d = [x, y, z, v_x, v_y, v_z]^T \tag{2}$$

with additional operational states such as radar mode, energy level, assigned task, and communication status.

5 Distributed Radar Network

5.1 Single Radar vs Distributed Radar

A major design decision was to prefer a distributed radar network over a single radar station. A single radar is easier to simulate but suffers from limited coverage, possible blind zones, and being a single point of failure. A distributed radar system, in contrast, offers:

- better coverage,
- better triangulation,

- improved robustness,
- fault tolerance,
- compatibility with swarm-based sensing.

5.2 Ground Radar and Drone-Mounted Radar

The final system uses two types of radar nodes:

- **Ground Radar:** static, long-range, broad area surveillance.
- **Drone Radar:** mobile, flexible, adaptive sensing, close-range geometry improvement.

This hybrid combination gives the system both stable global sensing and dynamic local refinement.

5.3 Radar Measurements

Each radar node can produce measurements such as:

$$z = [r, \theta, \phi, \dot{r}]^T \tag{3}$$

where:

- r is range,
- θ is azimuth angle,
- ϕ is elevation angle,
- $\dot{r}$ is radial velocity (Doppler).

In advanced modes, phase information may also be included.

6 Radar Signal Processing

6.1 FMCW Radar Concept

The simulator includes the idea of Frequency-Modulated Continuous Wave (FMCW) radar. In FMCW radar, the transmitted signal is a chirp whose frequency changes over time. Reflected signals are mixed with the transmitted waveform to produce a beat signal, from which range and velocity can be estimated.

6.2 Processing Pipeline

The classical radar signal chain discussed for RADAR-SWARM is:

1. transmit chirp,
2. receive echo,
3. beat signal generation,
4. range FFT,
5. Doppler FFT,
6. Range-Doppler map generation,
7. CFAR detection,
8. clustering of detections.

6.3 Range and Doppler Estimation

A simplified range estimate in FMCW radar is obtained from beat frequency f_b as:

$$R = \frac{cf_b}{2S} \tag{4}$$

where c is the speed of light and S is chirp slope.

Doppler shift gives radial velocity information, typically through slow-time FFT processing across chirps.

6.4 CFAR Detection

Constant False Alarm Rate (CFAR) detection was included as the main approach for adaptive thresholding in noisy radar data. CFAR compares the value of a cell under test with surrounding training cells and guard cells to determine whether a detection is significant.

7 Detection Clustering

Radar detections often contain multiple nearby points corresponding to the same physical object. To address this, clustering methods such as DBSCAN were discussed.

7.1 Why Clustering Matters

Without clustering, one target may appear as many independent detections. Clustering groups nearby detections and provides a centroid estimate for later tracking stages.

7.2 DBSCAN

DBSCAN was preferred because it can:

- identify dense groups,
- reject isolated noise points,
- operate without specifying the number of clusters in advance.

8 Multi-Sensor Fusion

8.1 Motivation

Since the radar network is distributed, each radar node sees the environment from a different angle. Fusion combines these measurements into a single coherent target estimate.

8.2 Fusion Goal

The aim is to estimate a global target position:

$$\hat{\mathbf{x}} = [x, y, z]^T \tag{5}$$

from multiple local radar measurements.

8.3 Benefits of Fusion

Multi-sensor fusion improves:

- accuracy,
- robustness,
- observability,
- resilience to sensor failure.

8.4 Centralized, Decentralized, and Hybrid Fusion

Three styles of fusion were explored:

- **Centralized fusion:** all measurements go to one global fusion server.
- **Decentralized fusion:** each drone maintains its own world model.
- **Hybrid fusion:** local fusion on drones plus periodic synchronization.

The hybrid model was recognized as especially attractive because it combines local autonomy with global consistency.

9 Multi-Target Tracking

9.1 Why Tracking Is Needed

Detections are noisy and incomplete. Tracking maintains continuity of target identity over time and predicts future target motion.

9.2 Track State

A track may be represented as:

$$\mathbf{x}_k = [x, y, z, v_x, v_y, v_z]^T \tag{6}$$

with an associated covariance matrix P_k measuring uncertainty.

9.3 Kalman Filtering

The Kalman filter was selected as a core tracking tool due to its efficiency and suitability for linear-Gaussian tracking. The two fundamental stages are:

$$\text{Prediction:} \quad \hat{\mathbf{x}}k|k-1 = F\hat{\mathbf{x}}k-1|k-1 \text{ Update:} \quad \hat{\mathbf{x}}k|k = \hat{\mathbf{x}}k|k-1 + K_k(z_k - H\hat{\mathbf{x}}_{k|k-1}) \tag{7}$$

where F is the state transition matrix, H is the measurement model, and K_k is the Kalman gain.

9.4 Data Association

Since multiple targets may be present, assigning detections to tracks becomes essential. Hungarian assignment was discussed for this purpose because it can efficiently solve a global association problem.

9.5 Multi-Hypothesis Tracking

Inspired partly by the idea of quantum superposition, the concept of maintaining multiple possible target trajectories before finalizing one was also discussed. This resembles multi-hypothesis tracking and is useful under ambiguity.

10 Drones as Mobile Radar Nodes and Interceptors

10.1 Dual Role of Drones

A defining feature of RADAR-SWARM is that drones are not just interceptors. They also act as mobile sensing nodes. This gives each drone two simultaneous possible roles:

- **Radar Node:** scanning and reporting target measurements.
- **Interceptor:** moving toward or around targets for pursuit and observation.

10.2 Why This Matters

Mobile radar drones allow the system to:

- reduce blind spots,
- reposition for better geometry,
- refine uncertain tracks,
- maintain surveillance in dynamic conditions.

11 Distributed Swarm Intelligence

11.1 Centralized vs Distributed Swarm Control

A key design choice was to prefer distributed swarm intelligence. In a centralized system, one controller decides everything. In a distributed system, drones make decisions locally while sharing information with neighbors.

The distributed model was favored because it offers:

- better scalability,
- better fault tolerance,
- more realistic swarm behavior,
- stronger research relevance.

11.2 Typical Drone Roles

Possible task roles include:

- interceptor,
- radar support node,
- communication relay,
- geometry optimizer,
- search agent.

12 Adaptive Radar Swarm Geometry

12.1 Core Idea

One of the strongest concepts developed for the project was **adaptive radar swarm geometry**. Instead of drones moving only to chase targets, they also move to improve sensor geometry for localization.

12.2 Why Geometry Matters

Localization accuracy depends strongly on the relative placement of sensors. If radar nodes are nearly collinear with the target, triangulation becomes weak. If they surround the target with good angular diversity, localization improves.

12.3 Geometric Principle

A simplified goal is to maximize angular separation between sensing nodes. Good geometry generally improves observability and reduces track covariance.

12.4 Connection to Information Gain

Adaptive geometry was later connected to information-theoretic planning, where drones are directed to positions that reduce target uncertainty the most.

13 Communication, Information Flow, and Bandwidth Management

13.1 Why Communication Is a Core Problem

In a distributed swarm, communication architecture is just as important as sensing and tracking. Each drone may produce large amounts of radar data, and indiscriminate data sharing would overload the network.

13.2 Levels of Data Sharing

Three levels of information sharing were distinguished:

1. **Raw radar data** — highest bandwidth cost.
2. **Detections** — moderate bandwidth cost.
3. **Tracks** — low bandwidth cost.

Track-to-track fusion was identified as particularly practical because it exchanges compact information.

13.3 Neighborhood Communication

Instead of broadcasting everything globally, local peer-to-peer communication was emphasized. This aligns naturally with decentralized swarm intelligence.

13.4 Priority-Based Messaging

Important messages such as imminent collisions or new critical target tracks should receive higher priority than routine telemetry.

14 Information Fields and Information Gain Planning

14.1 High-Level Idea

The airspace can be modeled as a map of uncertainty. Drones should move not only toward targets, but also toward locations where new measurements will reduce uncertainty the most.

14.2 Entropy as Uncertainty

Information theory measures uncertainty through entropy:

$$H(X) = -\sum_{x} p(x) \log p(x) \tag{8}$$

If a new observation reduces uncertainty, then information gain is:

$$I = H_{\text{before}} - H_{\text{after}} \tag{9}$$

14.3 Importance for Swarms

This allows drones to choose sensing positions rationally rather than randomly. It also unifies exploration, target tracking, and geometry optimization.

15 Global Objective Function of the Swarm

A major synthesis step in the design process was the introduction of a unified utility function:

$$U = \alpha I - \beta E - \gamma C - \delta R \tag{10}$$

where:

- I = information gain,
- E = energy consumption,
- C = communication cost,
- R = operational risk.

This objective formalizes the swarm's behavior as a trade-off between sensing quality and operational cost. Drones choose actions that maximize U.

16 Bio-Inspired Coordination Concepts

Nature-inspired ideas were introduced because biological swarms solve distributed coordination problems efficiently without central control.

16.1 Ant Colonies

Ants use pheromone trails, leading to the idea of digital information trails or coverage maps in the simulator.

16.2 Bird Flocks

Bird flocking inspired the use of local rules such as separation, alignment, and cohesion for stable motion and formation behavior.

16.3 Fish Schools

Fish schools demonstrated how fast local reactions can propagate through a group. This inspired local wave-like information propagation in the swarm.

16.4 Bees

Bee colonies inspired decentralized task allocation and consensus-like behavior.

16.5 Termites

Termites inspired self-organizing structures, suggesting that swarm formations could emerge from simple local rules.

17 Quantum-Inspired Concepts

A large part of the conceptual exploration involved identifying quantum phenomena that could be imitated algorithmically in a classical simulator.

17.1 Important Clarification

The project does **not** simulate real quantum hardware or full quantum mechanics. Instead, it uses **quantum-inspired ideas** as algorithmic metaphors or design principles.

17.2 Time Crystal and Time Quasicrystal Inspiration

Time crystals and time quasicrystals inspired the idea of temporal coordination. Instead of all drones acting every frame, drones and radar nodes may follow structured temporal patterns for:

- radar scans,
- communication bursts,
- navigation updates.

This reduces communication overload and creates synchronized swarm rhythms.

17.3 Entanglement-Inspired Coordination

The idea of quantum entanglement inspired tightly coupled drone pairs or groups whose states are strongly linked. This was interpreted as rapid shared-state coordination.

17.4 Superposition-Inspired Multi-Hypothesis Tracking

Quantum superposition inspired keeping multiple target trajectory hypotheses alive until enough evidence is gathered.

17.5 Quantum Annealing-Inspired Optimization

Quantum annealing inspired the use of optimization procedures that seek globally better sensing configurations, especially for geometry and task allocation.

17.6 Interference-Inspired Cooperative Sensing

Quantum and wave interference inspired the concept of multiple radar nodes cooperating so that signal patterns strengthen detection or improve sensing quality.

17.7 Tunneling-Inspired Escape from Local Optima

Quantum tunneling inspired random or probabilistic jumps in optimization, allowing the swarm to escape poor local configurations.

17.8 Measurement Collapse and Decision Finalization

The idea of wavefunction collapse inspired final decision commitment after maintaining uncertainty or multiple hypotheses during earlier stages.

18 Quantum Zeno Effect, Topological Protection, and Quantum Walks

18.1 Quantum Zeno Effect

The quantum Zeno effect inspired a strategy where rapid repeated measurements temporarily stabilize an uncertain track. In the simulator, if target uncertainty grows too much, drones may increase sensing frequency to reduce divergence.

18.2 Topological Protection

Topological protection inspired robust communication architectures in which drone communication graphs remain resilient to failures, often by maintaining loops or redundant paths.

18.3 Quantum Walks

Quantum walks inspired exploration algorithms that spread more efficiently through the search space than simple random motion. This connects naturally to information-driven exploration.

19 Berry Phase, Teleportation, and Adiabatic Evolution

19.1 Berry Phase

Berry phase inspired geometry-aware formation control, especially orbiting surveillance around a target where path geometry matters, not just start and end points.

19.2 Teleportation-Inspired State Transfer

Quantum teleportation inspired the idea of track state handoff between drones. A drone leaving coverage can transfer not just a target detection but an entire tracking state estimate to another drone.

19.3 Adiabatic Evolution

Adiabatic evolution inspired smooth swarm reconfiguration. Instead of sudden role changes or large movement jumps, the swarm can gradually adapt while preserving stability.

20 Squeezed Sensing Modes

20.1 Physical Inspiration

Quantum squeezed states reduce uncertainty in one variable while increasing it in another. This inspired adaptive measurement modes in which drones prioritize range accuracy, velocity accuracy, or angular accuracy depending on the task.

20.2 Covariance Interpretation

In a simplified model, the sensor noise covariance matrix can be reshaped so that one dimension becomes more precise while another becomes noisier. Sensor fusion then combines complementary measurements from multiple drones.

21 Correlation Radar and Quantum Radar Inspiration

21.1 Quantum Radar Inspiration

Although real quantum radar remains highly experimental, the project borrowed the idea of correlation-based detection from quantum illumination.

21.2 Classical Simulation Interpretation

A classical approximation is to transmit known coded signals and compare returns against stored references. The key detection metric becomes cross-correlation:

$$C = \langle r(t), s(t) \rangle \tag{11}$$

where $r(t)$ is received signal and $s(t)$ is reference waveform.

21.3 Importance for Weak Targets

Correlation-based detection is especially attractive for weak targets in noisy conditions because detection relies on pattern matching rather than signal power alone.

22 Distributed Radar Interferometry

22.1 Core Principle

Interferometry uses phase differences between radar nodes to obtain very fine localization information.

If two sensors receive the same reflected wave, the phase difference can be related to path length difference:

$$\Delta d = \frac{\lambda}{2\pi} \Delta \phi \tag{12}$$

where λ is wavelength and $\Delta\phi$ is phase difference.

22.2 Why It Matters

This enables super-resolution localization beyond what conventional range-only measurements would easily provide.

22.3 Role of Swarm Geometry

Interferometric performance benefits from carefully chosen baselines between drone radar nodes, further strengthening the case for adaptive radar swarm geometry.

23 Synthetic Aperture Radar (SAR) Swarm

23.1 SAR Principle

Synthetic aperture radar uses motion to create the equivalent of a larger antenna. By combining measurements from many positions over time, spatial resolution improves dramatically.

23.2 Swarm Extension

In RADAR-SWARM, multiple moving drones can jointly form a synthetic aperture. This transforms the system from pure tracking toward radar imaging.

23.3 Potential Capabilities

A SAR swarm could allow:

- higher target resolution,
- shape estimation,
- better classification,
- target imaging instead of only point tracking.

24 Electronic Deception and EM-Based Misguidance

24.1 Important Physical Clarification

A question was raised about whether mobile radar drones could manipulate electromagnetic field strength to physically misguide a target. The conclusion was that directly pushing or steering neutral aerial targets with radar EM force is not realistic.

24.2 Realistic Interpretation

A much more realistic interpretation is electronic deception:

- radar jamming,
- spoofing,
- decoys,
- false target generation,
- navigation deception.

Within the simulator, this would mean manipulating the target's perceived sensing environment rather than physically moving it.

25 Python and C++ Segregation

25.1 Why a Hybrid Language Architecture Was Proposed

Since RADAR-SWARM includes both high-level orchestration and computationally heavy processing, a hybrid architecture using Python and C++ was proposed.

25.2 Suggested Split

Python should handle:

- main simulation loop,
- swarm logic,
- task allocation,

- temporal coordination,
- visualization.

  C++ should handle:

- radar signal processing,
- FFT-heavy operations,
- CFAR detection,
- tracking kernels,
- performance-critical fusion code.

25.3 Bindings

Modern Python bindings such as `pybind11` were suggested as a good interface between the two languages.

26 Execution Loop of the Simulator

The core simulation loop discussed for RADAR-SWARM is:

```
This loop represents the heartbeat of the entire platform.

\section{Unified Layered Architecture} A coherent layered architecture emerged from
    the discussion: \begin{enumerate}[leftmargin=1.5em] \item Environment layer, \item
     Distributed radar layer, \item Signal processing layer, \item Fusion layer, \item
     Tracking layer, \item Communication layer, \item Swarm intelligence layer, \item
    Quantum-inspired and bio-inspired coordination layer, \item Imaging and
    classification layer. \end{enumerate}

This architecture helps prevent the project from becoming a disconnected set of
    features.

\section{Minimal Realistic Development Strategy} Because the full concept is highly
    ambitious, an incremental implementation strategy is recommended: \subsection{
    Phase 1} 3D environment, one radar, simple targets, basic tracking.

\subsection{Phase 2} Distributed radar nodes, multi-sensor fusion, improved tracking.

\subsection{Phase 3} Drone-mounted radar nodes, swarm behavior, adaptive geometry.

\subsection{Phase 4} Communication constraints, information gain planning,
    decentralized coordination.

\subsection{Phase 5} Advanced sensing such as interferometry, correlation-based
    detection, SAR, and quantum-inspired enhancements.

\section{One-Sentence Summary of the Project} RADAR-SWARM is a \textbf{3D simulation
    platform for autonomous drone swarms that cooperatively sense, track, and analyze
    aerial targets using distributed radar networks, adaptive geometry, information-
    driven control, and quantum-inspired coordination concepts.}
```